

Rafi Mahmud Tumon

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Education

Technical University Berlin	Berlin, Germany
<i>Master of Science in Geodesy and Geoinformation Science</i>	<i>October 2020 – Present</i>
• Specialization: Geoinformation Science; Key Coursework: Photogrammetric Computer Vision, Geoinformatics, Geodatabases and Infrastructures, Space Geodesy, Adjustment Theory, Deep Learning for Geographic Data	
Rajshahi University of Engineering and Technology	Rajshahi, Bangladesh
<i>Bachelor of Science in Urban and Regional Planning</i>	<i>March 2014 – November 2018</i>

Work Experience

German Aerospace Center (DLR)	Berlin, Germany
<i>Student Assistant</i>	<i>January 2026 – Present</i>
• Application of machine learning methods with multimodal geodata in urban areas. • Utilize Python and QGIS for geospatial analysis and model development.	
OXG Glasfaser GmbH	Berlin, Germany
<i>Working Student, Strategic Area Development</i>	<i>June 2024 – Present</i>
• Improved operational efficiency by 20% through automated geospatial analysis using QGIS and Python for fiber network deployment. • Conduct cost-benefit analyses and site assessments to identify optimal deployment areas while navigating geographical constraints and market competition. • Developed Python-based geocoding tool for German addresses, reducing processing time by 30%.	
Department of Civil Engineering, RUET	Rajshahi, Bangladesh
<i>Project Assistant (GIS and Geospatial Planner)</i>	<i>February 2020 – April 2020</i>
• Led an airport master planning project using GIS analysis to improve regional connectivity for underserved communities in northwestern Bangladesh. • Collaborated with engineering teams and stakeholders to integrate spatial analysis into transportation planning decisions.	
Urban Development Directorate	Rajshahi, Bangladesh
<i>GIS Analyst</i>	<i>December 2018 – January 2020</i>
• Conducted socio-economic surveys and spatial analysis for an urban planning project in Bagha Upazilla, collecting field data from local communities. • Maintained geodatabases tracking urbanization indicators to support resource allocation decisions for deprived areas.	

Research Experience

Multimodal Deep Learning for Urban Tree Crown Condition Assessment	<i>Present</i>
<i>Master's Thesis: "Multimodal Stereo Image Fusion for Urban Tree Vitality Classification Using Deep Learning"</i>	
• Implemented YOLOv12s-based tree crown detection on large-format GeoTIFF imagery for dataset preparation. • Constructed geospatial preprocessing pipelines for spatial joins, CRS-consistent raster cropping, and multi-channel patch extraction. • Designing a PyTorch Lightning classification framework for tree vitality assessment. • Evaluating encoder architectures and multimodal fusion strategies to determine optimal configurations.	
Flood Vulnerability Assessment Using Geospatial Analysis	<i>November 2018</i>
<i>Bachelor's Thesis: "Application of Remote Sensing and GIS for Flood Inundation Mapping and Vulnerability Assessment: A Case Study of Bogra, Sirajganj, and Tangail Districts"</i>	
• Conducted vulnerability assessment for three flood-prone districts in Bangladesh using satellite imagery to identify at-risk communities. • Developed geospatial methodologies and produced flood risk maps to support disaster planning.	

Personal Projects

Forest Loss Analysis in Colombia's Indigenous Lands

github.com/rafimt/Deforestation-Analysis

- Analyzed 10 years of satellite images using Google Earth Engine and Python to track deforestation and its effects on vulnerable communities, revealing a 15% loss in tree cover.

Wildfire Data Analysis

github.com/rafimt/Wildfire_Analysis_2022

- Developed a Python tool to analyze wildfire patterns in 8 countries (analyzing 1,234 fires total) to help with disaster planning.

Berlin's Environmental Dynamics Analysis

github.com/rafimt/Project-Geoinformatics-2024

- Assessed Urban Heat Island effects using satellite data, finding that built-up areas experienced the highest temperatures compared to other land types.

Skills and Tools

- **Geospatial & Remote Sensing:** QGIS (Data Management, Coordinate Reference Systems, Map Creation and Cartography, Spatial Joins, Processing Modeler, Python Scripting and Workflow Automation), Google Earth Engine, PostgreSQL/PostGIS, GeoPandas, Rasterio, OSMnx, GDAL, GNSS Processing, Least Squares Adjustment, Cartographic Principles
- **Machine Learning & Deep Learning:** PyTorch, Scikit-learn, Object Detection (YOLO), CNNs, Vision Transformers (ViT), Graph Neural Networks, Focal Loss, Model Evaluation & Classification
- **Programming & Development:** Python (GeoPandas, Matplotlib, Seaborn, Rasterio, geemap, SAHI, OSMnx), SQL, Git, Jupyter Notebook
- **Domain Expertise:** Spatial Data Management, Environmental Analysis, Vulnerability Assessment, Engineering Survey, Urban & Regional Planning, Field Research & Data Collection

Awards and Recognition

Planning and Design of Compact Township, 2nd Runner Up

Inter-University Undergraduate Planning Students Competition, Bangladesh University of Engineering and Technology (BUET)

Publication

"Identification of Determinant Factors Influencing Modal Choice Behavior and Satisfaction Level of Commuters: A Case of Rajshahi City"

Trends in Civil Engineering and its Architecture, Lupine Publishers

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Languages

English (Professional working proficiency), **German** (Elementary proficiency - A2)